

Temporal Structure in the Yeast Transcriptome

Period Discovery, Gene Clustering, and Semantic Annotation

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Outline

The Yeast Metabolic Cycle

- *Saccharomyces cerevisiae* (budding yeast) exhibits a spontaneous, synchronized **metabolic oscillation** when grown in continuous culture.
- Thousands of genes are expressed periodically, in groups called **temporal modules**

Two Central Questions

1. What is the period of the metabolic cycle, and what are its phases?
2. Which genes drive each phase, and what biological processes do they represent?

Dataset (maybe replace this slide with diagram)

- **Source:** Tu et al. (2005), *Science* — publicly available via GEO (GSM77298).
- **Assay:** Yeast whole-transcriptome microarray.
- **Features:** 9,725 RNA transcripts.
- **Time points:** 36 samples at ≈ 25 -minute intervals (≈ 15 hours total).
- Each time point is an independent microarray chip measuring relative transcript abundance across the *same* oscillating population.

Preprocessing Pipeline

Raw intensities require three normalization steps before analysis.

1. **Per-chip median normalization.** Each chip is divided by its median intensity, removing global staining and loading differences between arrays.
 - Assumption: true median expression is constant across time.
 - Result: *relative* (not absolute) expression.
2. **arcsinh transform. (maybe a graph would be good here)** Compresses the dynamic range (similar to log, but handles zeros).

$$\tilde{x} = \operatorname{arcsinh}(x) = \ln(x + \sqrt{x^2 + 1})$$

3. **Per-gene z -score** (for clustering only). Each gene's time series is standardized to zero mean and unit variance so genes with very different baseline expressions are comparable.

Approach: Clustering Time Points

Key Insight

If the metabolic cycle has period T , time points separated by T should look alike — they should fall in the *same cluster*.

Strategy: Cluster the 36 *time points* (not genes) using a Gaussian Mixture Model (GMM), then read off the period from the repeating cluster sequence.

Dimensionality reduction (necessary — $n = 36 \ll d = 9725$):

- Approach 1: Keep top- n highest-variance genes, sweep $n \in [200, 1100]$.
- Approach 2: Project all genes to 5 PCA components; select k by held-out log-likelihood (first 24 train, last 12 test).
- Both approaches recover $k = 4$ clusters.

Result: Period = 12 Time Steps, 300 seconds

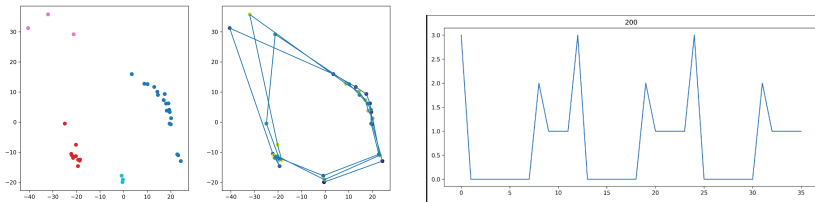


Figure: Left: 36 time points in PCA space, colored by GMM cluster. Right: Cluster assignment vs. time step, showing three complete repetitions of the 4-phase cycle.

Period Discovery — Summary

- Every dimensionality-reduction and GMM variant converged to a **12-timestep period**.

$$T = 12 \times 25 \text{ min} = 300 \text{ min}$$

- Cluster assignments are near-deterministic (high posterior probabilities), indicating well-separated phases. Also curse of dimensionality.
- The 36-point experiment spans exactly **3 complete cycles**.
- Independent confirmation of the period reported by Tu et al.

Identifying Periodic Genes

With period $T = 12$ established, which genes actually oscillate?

Permutation test on lag-12 autocorrelation:

1. Compute $\text{ACF}(x_g, \tau = 12)$ for each gene g .
2. Shuffle gene g 's time series 100 times; compute ACF for each shuffle.
3. Accept gene as periodic if its true ACF is at or above the **95th percentile** of the shuffled distribution.

Result: 5,243 of 9,725 genes ($\approx 54\%$) are identified as periodic at lag 12.

Periodic vs. Non-Periodic Genes in PCA

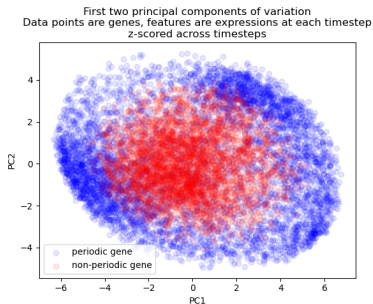
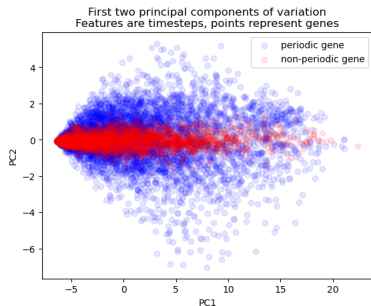


Figure: PCA of all 9,725 genes across 36 timesteps. Left: arcsinh expression (PC1 captures mean expression level). Right: After per-gene z -score normalization, periodic and non-periodic genes separate clearly in both PCs.

Classical Time Series Decomposition

For each periodic gene, decompose its standardized time series:

$$x_g(t) = \underbrace{m_g(t)}_{\text{trend}} + \underbrace{s_g(t)}_{\text{seasonal}} + \underbrace{\varepsilon_g(t)}_{\text{residual}}$$

Trend estimated via centered moving average (24 interior time points).

Clustering the seasonal components:

- Fit GMMs over $k = 2, \dots, 12$; select k by AIC/BIC.
- $k = 8$ **minimizes BIC** and gives a local AIC minimum.
- 7 clusters show distinct temporal expression profiles; 1 cluster is a catch-all for noisy genes.

Temporal Expression Modules

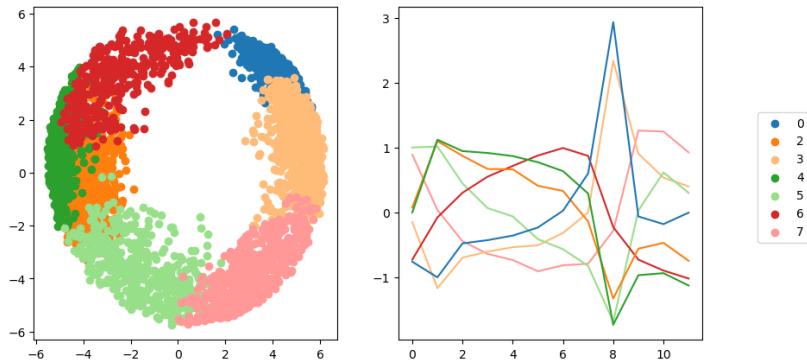


Figure: Left: UMAP of genes by seasonal component, colored by GMM cluster. Right: Mean seasonal profile for each of the 7 coherent clusters, showing peaks and troughs at distinct phases of the 12-step cycle.

ARMA Forecasting on Individual Genes

- ARMA(p, q) models fit to each periodic gene's time series.
- **Train:** first 24 time steps (2 full cycles). **Test:** last 12 time steps (1 held-out cycle).
- Order (p, q) selected per gene by AIC.
- Models capture the periodic structure and **predict the final cycle** with reasonable accuracy for well-expressed, high-amplitude genes.
- Performance degrades for low-amplitude or noisy genes — motivating more expressive sequence models.

Continuous-Density HMM Prediction

- A **Gaussian mixture HMM (CDHMM)** treats the hidden metabolic phase as a latent Markov chain and models observations with a GMM emission distribution.
- Hidden states correspond to the 4 metabolic phases recovered earlier.
- Trained on the first 24 time steps; evaluated on the final 12.
- Captures **phase transitions** explicitly — emission distributions encode which genes are up/down in each phase.
- Prediction: given observed gene expression, infer the current phase and project forward.

What Are the Gene Clusters *Doing*?

Clustering recovers *when* groups of genes are expressed. To understand *why*, we need biological annotation.

Approach: Latent Dirichlet Allocation (LDA) on gene-level metadata (GO terms, functional annotations).

- Each gene is a “document”; its annotations are “words”.
- LDA discovers $K = 10$ latent **functional topics**.
- Topic weights per gene are then correlated with the time-series clusters to ask: *which biological processes dominate each temporal module?*

LDA Topics (labeled by Claude)

Topic	Short Name
1	Cytoplasmic Translation & Ribosome Structure
2	Cell Cycle & DNA Damage Checkpoint Kinases
3	Proteasome & ATP-Dependent Protein Degradation
4	RNA Pol II Transcription & Chromatin Remodeling
5	ER Membrane & Transmembrane Proteins
6	Mitotic/Meiotic Chromosome Segregation
7	Mitochondrial Metabolism & Oxidative Phosphorylation
8	Vesicular Transport & Endomembrane System
9	Retrotransposon Biology (Ty Elements)
10	Ribosome Biogenesis & rRNA Processing

Topic-Cluster Associations

Figure: Heatmap of LDA topic weight vs. time-series cluster. Each cell shows the fraction of genes in a cluster assigned to a given functional topic, revealing which biological processes are co-activated in each temporal module.

Key Biological Findings (This was written by Claude, check)

Combining the temporal clusters with LDA topic weights reveals:

- **Early cycle (oxidative phase):** enrichment for *mitochondrial metabolism & oxidative phosphorylation* (Topic 7).
- **Mid cycle:** enrichment for *ribosome biogenesis* (Topic 10) and *cytoplasmic translation* (Topic 1) — cells ramping up biosynthetic capacity.
- **Late cycle (reductive phase):** *cell cycle checkpoints* (Topic 2) and *chromosome segregation* (Topic 6) — consistent with DNA replication gated to this phase.
- *Proteasome* activity (Topic 3) appears across multiple phases, consistent with ongoing protein quality control.

Summary

1. **Period:** GMM clustering of time points robustly recovers $T = 12$ steps = 300 minutes across all dimensionality-reduction strategies.
2. **Periodic genes:** 5,243/9,725 genes ($\approx 54\%$) oscillate with this period (permutation test, $\alpha = 0.05$).
3. **Temporal modules:** Seasonal decomposition + GMM identifies 7 coherent clusters of co-expressed genes, each with a distinct phase profile.
4. **Forecasting:** ARMA and CDHMM models capture the periodic dynamics and predict held-out expression.
5. **Function:** LDA links each temporal module to specific biological processes — oxidative metabolism, translation, replication, degradation — giving a functional portrait of the yeast metabolic cycle.

Thank You

Questions?

Data: Tu et al. (2005) *Science*, GEO accession GSM77298.

Code: https://github.com/jackdpond/hmm_gene_model

Backup: GMM Model Selection

- Held-out log-likelihood evaluated over 100 random restarts for $k \in \{2, \dots, 7\}$.
- $k = 4$ consistently maximized held-out likelihood.
- Tied covariance used (full covariance is ill-conditioned for $n = 24, d = 5$).

Backup: Permutation Test Details

For each gene g with time series $\mathbf{x}_g$:

$$r_g = \frac{1}{n - \tau} \sum_{t=1}^{n-\tau} \frac{(x_g(t) - \bar{x}_g)(x_g(t + \tau) - \bar{x}_g)}{s_g^2}, \quad \tau = 12$$

Null distribution: 100 random permutations of $\mathbf{x}_g$.

Reject null if $r_g \geq$ 95th percentile of permuted $r_g^{(b)}$.

Backup: LDA Generative Model

For each gene g :

1. Draw topic proportions $\theta_g \sim \text{Dir}(\alpha)$.
2. For each annotation term w :
 - 2.1 Draw topic $z \sim \text{Categorical}(\theta_g)$.
 - 2.2 Draw term $w \sim \text{Categorical}(\phi_z)$.

Inference via variational EM; $K = 10$ topics selected by held-out perplexity. Topic names assigned by prompting Claude with top-20 terms per topic.